PCM INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

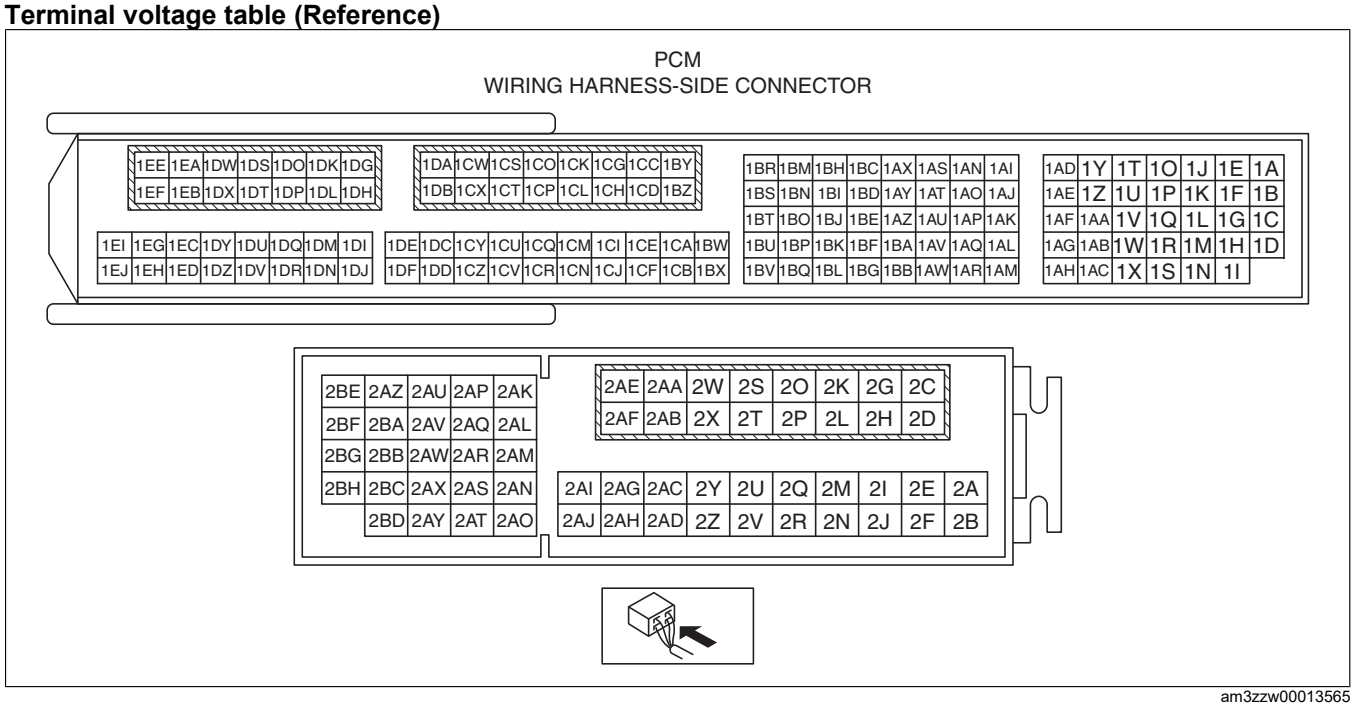
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Without Using the M-MDS

Note

- Because the PCM uses a waterproof connector, the inspection for the voltage/wave pattern cannot be performed. The following values are for reference.

Terminal voltage table (Reference)



am3zzw00013565

—: Not applicable

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1A*1	CAN2_H	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
1B*1	CAN2_L	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
1C	—	—	—		—	—
1D	Knocking (–)	KS	Ignition switched ON (engine off) (Use digital type voltmeter, because measurement voltage will be detected less than true voltage when using analog type voltmeter)		Approx. 1.64	• KS • Related wiring harness
1E	—	—	—		—	—
1F	—	—	—		—	—
1G*7	Neutral switch No.2	Neutral switch No. 2	Switch ignition ON (engine off)	Neutral	Below 1.0	• Neutral switch No.2 • Related wiring harness
				Except above	B+	
1H	Knocking (+)	KS	Ignition switched ON (engine off) (Use digital type voltmeter, because measurement voltage will be detected less than true voltage when using analog type voltmeter)		Approx. 3.28	• KS • Related wiring harness
1I	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
1J*5	Electric variable valve timing motor (rotation direction)	Electric variable valve timing motor/ driver	(See Electric variable valve timing motor (rotation direction) signal.)			• Electric variable valve timing motor/driver • Related wiring harness

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1K*2	Neutral position	Neutral switch No. 1	Shift lever is at neutral position		Below 1.0	• Neutral switch No.1 • Related wiring harness
			Shift lever is not at neutral position		B+	
1L*2	Back-up light	Back-up light switch	Shift lever is at R position		Below 1.0	• Back-up light switch • Related wiring harness
			Shift lever is not at R position		B+	
1M	—	—	—		—	—
1N	—	—	—		—	—
1O*5	Electric variable valve timing motor (rotation pulse)	Electric variable valve timing motor/ driver	(See Electric variable valve timing motor (rotation pulse) signal.)			• Electric variable valve timing motor/driver • Related wiring harness
1P	Oil pressure	Oil pressure switch	Ignition switched ON (engine off)		Below 1.0	• Oil pressure switch • Related wiring harness
			Idle (after warm up)		B+	
1Q	—	—	—		—	—
1R	—	—	—		—	—
1S	—	—	—		—	—
1T*5	Exhaust CMP	Exhaust CMP sensor	(See Exhaust CMP signal.)			• Exhaust CMP sensor • Related wiring harness
1U	—	—	—		—	—
1V	—	—	—		—	—
1W	A/F	A/F sensor	Idle (after warm up)		Approx. 4.18	• A/F sensor • Related wiring harness
1X*5	GND	Exhaust CMP sensor	Under any condition		Below 1.0	• Related wiring harness
1Y	Intake CMP	Intake CMP sensor	(See Intake CMP signal.)			• Intake CMP sensor • Related wiring harness
1Z*3	Generator output voltage	Generator	(See Generator output voltage (with i-ELOOP).)			• Generator • Related wiring harness
1AA	—	—	—		—	—
1AB	A/F	A/F sensor	Idle (after warm up): 0 mA			• A/F sensor • Related wiring harness
1AC	GND	Intake CMP sensor	Under any condition		Below 1.0	• Related wiring harness
1AD	CKP	CKP sensor	(See CKP signal.)			• CKP sensor • Related wiring harness
1AE*5	Electric variable valve timing driver (diagnostic)	Electric variable valve timing motor/ driver	(See Electric variable valve timing driver (diagnostic) signal.)			• Electric variable valve timing motor/driver • Related wiring harness
1AF*4	Generator output voltage	Generator	(See Generator output voltage (without i-ELOOP).)			• Generator • Related wiring harness
1AG	A/F	A/F sensor	Idle (after warm up)		Approx. 4.01	• A/F sensor • Related wiring harness
1AH	GND	CKP sensor	Under any condition		Below 1.0	• Related wiring harness
1AI	Purge control	Purge solenoid valve	(See Purge control.)			• Purge solenoid valve • Related wiring harness
1AJ	IGT4	Ignition coil No.4	(See IGT1, IGT2, IGT3, IGT4 control.)			• Ignition coil No.4 • Related wiring harness
1AK	ECT	ECT sensor	Ignition switched ON (engine off)	ECT 20 °C {68 °F}	Approx. 3.10	• ECT sensor • Related wiring harness
				ECT 40 °C {104 °F}	Approx. 2.16	
				ECT 60 °C {140 °F}	Approx. 1.40	
				ECT 80 °C {176 °F}	Approx. 0.87	
				ECT 100 °C {212 °F}	Approx. 0.54	
1AL	—	—	—		—	—
1AM	GND	ECT sensor	Under any condition		Below 1.0	• Related wiring harness
1AN	Hydraulic variable valve timing control	OCV	(See Hydraulic variable valve timing control signal.)			• OCV • Related wiring harness
1AO	IGT3	Ignition coil No.3	(See IGT1, IGT2, IGT3, IGT4 control.)			• Ignition coil No.3 • Related wiring harness
1AP	—	—	—		—	—

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1AQ	—	—	—		—	—
1AR	—	—	—		—	—
1AS *5	Engine oil control	Engine oil solenoid valve	(See Engine oil control signal.)			• Engine oil solenoid valve • Related wiring harness
1AT	IGT2	Ignition coil No.2	(See IGT1, IGT2, IGT3, IGT4 control.)			• Ignition coil No.2 • Related wiring harness
1AU	—	—	—		—	—
1AV*5	Ion (No.4)	Ion sensor No.4	Idle (after warm up)		Approx. 4.48	• Ion sensor No.4 • Related wiring harness
1AW	—	—	—		—	—
1AX	—	—	—		—	—
1AY	IGT1	Ignition coil No.1	(See IGT1, IGT2, IGT3, IGT4 control.)			• Ignition coil No.1 • Related wiring harness
1AZ*5	Electric variable valve timing control	Electric variable valve timing motor/ driver	(See Electric variable valve timing control signal.)			• Electric variable valve timing motor/driver • Related wiring harness
1BA*5	Ion (No.3)	Ion sensor No.3	Idle (after warm up)		Approx. 4.48	• Ion sensor No.3 • Related wiring harness
1BB	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
1BC	—	—	—		—	—
1BD	—	—	—		—	—
1BE*4	Generator field coil control	Generator	(See Generator field coil control signal (without i-ELOOP).)			• Generator • Related wiring harness
1BF*5	Ion (No.2)	Ion sensor No.2	Idle (after warm up)		Approx. 4.48	• Ion sensor No.2 • Related wiring harness
1BG*5	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
1BH	—	—	—		—	—
1BI	—	—	—		—	—
1BJ	Constant voltage (Vref)	Fuel pressure sensor	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
1BK*5	Ion (No.1)	Ion sensor No.1	Idle (after warm up)		Approx. 4.48	• Ion sensor No.1 • Related wiring harness
1BL	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
1BM	—	—	—		—	—
1BN	Constant voltage (Vref)	CKP sensor	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
1BO	Constant voltage (Vref)	MAP sensor	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
1BP	TP (No.1)	TP sensor No.1	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 1.15	• TP sensor No.1 • Related wiring harness
				Accelerator pedal fully depressed	Approx. 4.63	
1BQ	GND	TP sensor No.1, TP sensor No.2	Under any condition		Below 1.0	• Related wiring harness
1BR*3	Generator field coil control	Generator	(See Generator field coil control signal (with i-ELOOP).)			• Generator • Related wiring harness
1BS	Constant voltage (Vref)	TP sensor No.1, TP sensor No.2	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
1BT	—	—	—		—	—
1BU	TP (No.2)	TP sensor No.2	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 3.95	• TP sensor No.2 • Related wiring harness
				Accelerator pedal fully depressed	Approx. 0.45	
1BV	—	—	—		—	—

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1BW	MAP	MAP sensor	Ignition switched ON (engine off)		Approx. 4.15	• MAP sensor • Related wiring harness
			Idle (after warm up)		Approx. 1.96	
			Racing (Engine speed: 2,000 rpm)		Approx. 1.33	
1BX	GND	MAP sensor, IAT sensor No.2	Under any condition		Below 1.0	• Related wiring harness
1BY	A/F sensor heater control	A/F sensor heater	(See A/F sensor heater control signal.)			• A/F sensor heater • Related wiring harness
1BZ	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1CA	Fuel pressure	Fuel pressure sensor	Ignition switched ON (engine off)		Approx. 1.51	• Fuel pressure sensor • Related wiring harness
			Idle (after warm up)		Approx. 1.13	
1CB	GND	Fuel pressure sensor	Under any condition		Below 1.0	• Related wiring harness
1CC	Drive-by-wire control (–)	Throttle valve actuator	**Idle (after warm up)** Because the drive-by-wire control (-) terminal value varies depending on the vehicle, examination using only the ICC terminal is not possible. When performing the inspection, perform it together with the ICG terminal. • Type A — B+ • Type B — Approx. 0			• Throttle valve actuator • Related wiring harness
1CD	—	—	—		—	—
1CE	IAT (No.2)	IAT sensor No.2	Ignition switched ON (engine off)	IAT 20 °C {68 °F}	Approx. 3.57	• IAT sensor No.2 • Related wiring harness
				IAT 40 °C {104 °F}	Approx. 2.70	
				IAT 60 °C {140 °F}	Approx. 1.87	
1CF	—	—	—		—	—
1CG	Drive-by-wire control (+)	Throttle valve actuator	(See Drive-by-wire control (+) signal.)			• Throttle valve actuator • Related wiring harness
1CH	—	—	—		—	—
1CI	—	—	—		—	—
1CJ	—	—	—		—	—
1CK	Battery voltage	Main relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1CL	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1CM	—	—	—		—	—
1CN	—	—	—		—	—
1CO	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1CP	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1CQ	—	—	—		—	—
1CR	—	—	—		—	—
1CS	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1CT	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1CU	—	—	—		—	—
1CV	—	—	—		—	—
1CW	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1CX	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1CY	—	—	—		—	—
1CZ	—	—	—		—	—
1DA	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1DB	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1DC	—	—	—		—	—
1DD	—	—	—		—	—
1DE	—	—	—		—	—

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
1DF	—	—	—		—	—
1DG	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1DH	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1DI	—	—	—		—	—
1DJ	—	—	—		—	—
1DK	Battery voltage	Fuel injector relay	Ignition switched ON (engine off)		B+	• Related wiring harness
1DL	GND	GND	Under any condition		Below 1.0	• Related wiring harness
1DM	—	—	—		—	—
1DN	—	—	—		—	—
1DO	Fuel injection control (–)	Fuel injector No.1	(See Fuel injection control (-) signal.)			• Fuel injector No.1 • Related wiring harness
1DP	Fuel injection control (+)	Fuel injector No.1	(See Fuel injection control (+) signal.)			• Fuel injector No.1 • Related wiring harness
1DQ	—	—	—		—	—
1DR	—	—	—		—	—
1DS	Fuel injection control (–)	Fuel injector No.4	(See Fuel injection control (-) signal.)			• Fuel injector No.4 • Related wiring harness
1DT	Fuel injection control (+)	Fuel injector No.4	(See Fuel injection control (+) signal.)			• Fuel injector No.4 • Related wiring harness
1DU	—	—	—		—	—
1DV	—	—	—		—	—
1DW	Fuel injection control (–)	Fuel injector No.2	(See Fuel injection control (-) signal.)			• Fuel injector No.2 • Related wiring harness
1DX	Fuel injection control (+)	Fuel injector No.2	(See Fuel injection control (+) signal.)			• Fuel injector No.2 • Related wiring harness
1DY	—	—	—		—	—
1DZ	—	—	—		—	—
1EA	Fuel injection control (–)	Fuel injector No.3	(See Fuel injection control (-) signal.)			• Fuel injector No.3 • Related wiring harness
1EB	Fuel injection control (+)	Fuel injector No.3	(See Fuel injection control (+) signal.)			• Fuel injector No.3 • Related wiring harness
1EC	—	—	—		—	—
1ED	—	—	—		—	—
1EE	High pressure fuel pump control (+)	High pressure fuel pump	(See High pressure fuel pump control (+) signal.)			• High pressure fuel pump • Related wiring harness
1EF	High pressure fuel pump control (–)	High pressure fuel pump	(See High pressure fuel pump control (-) signal.)			• High pressure fuel pump • Related wiring harness
1EG	—	—	—		—	—
1EH	—	—	—		—	—
1EI	—	—	—		—	—
1EJ	—	—	—		—	—
2A	—	—	—		—	—
2B	—	—	—		—	—
2C	HO2S heater control	HO2S heater	(See HO2S heater control signal.)			• HO2S heater • Related wiring harness
2D	—	—	—		—	—
2E	—	—	—		—	—
2F	—	—	—		—	—
2G	Brake (No.1)	Brake switch (No.1 signal)	Brake pedal released		Below 1.0	• Brake switch (No.1 signal) • Related wiring harness
			Brake pedal fully depressed		B+	
2H	Ignition (IG1)	IG1 relay	Ignition switched ON (engine off)		B+	• IG1 relay • Related wiring harness
2I	Ambient temperature	Ambient temperature sensor	Ignition switched ON (engine off)	AAT 20 °C {68 °F}	Approx. 2.70	• Ambient temperature sensor • Related wiring harness
				AAT 30 °C {104 °F}	Approx. 1.80	
2J*2	CPP	CPP switch, start stop unit	Clutch pedal fully depressed		Below 1.0	• CPP switch • Start stop unit • Related wiring harness
			Clutch pedal released		B+	

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
2K	Main relay control	Main relay	Ignition switched ON (engine off)		Approx. 0.83	• Main relay • Related wiring harness
2L	—	—	—		—	—
2M*7	Clutch stroke sensor	Clutch stroke sensor	Ignition switched ON (engine off)	Clutch pedal released	Approx. 4.56	• Clutch stroke sensor • Related wiring harness
				Clutch pedal fully depressed	Approx. 1.06	
2N	—	—	—		—	—
2O	Battery voltage	Battery	Under any condition		B+	• Battery • Related wiring harness
2P	DC-DC converter control*8	DC-DC converter	Ignition switched ON (engine off)		Below 1.0	• DC-DC converter • Related wiring harness
	DC-DC converter control*3	DC-DC converter (i-ELOOP)	Ignition switched ON (engine off)		Below 1.0	• DC-DC converter (i-ELOOP) • Related wiring harness
2Q*6	Power brake unit vacuum	Power brake unit vacuum sensor	Idle (after warm up)	Brake pedal released	Approx. 0.47	• Power brake unit vacuum sensor • Related wiring harness
2R	Brake (No.2)	Brake switch (No.2 signal)	Brake pedal released		Below 1.0	• Brake switch (No.2 signal) • Related wiring harness
			Brake pedal fully depressed		B+	
2S	Battery voltage	Main relay	Ignition switched ON (engine off)		B+	• Related wiring harness
2T	Battery voltage	Main relay	Ignition switched ON (engine off)		B+	• Related wiring harness
2U	IAT (No.1)	IAT sensor No.1	Ignition switched ON (engine off)	IAT 20 °C {68 °F}	Approx. 2.64	• IAT sensor No.1 • Related wiring harness
				IAT 40 °C {104 °F}	Approx. 1.27	
				IAT 60 °C {140 °F}	Approx. 0.66	
2V	—	—	—		—	—
2W*3	DC-DC converter (i-ELOOP) control	DC-DC converter (i-ELOOP)	Idle (after warm up)		0 or B+	• DC-DC converter (i-ELOOP) • Related wiring harness
2X	—	—	—		—	—
2Y	—	—	—		—	—
2Z	—	—	—		—	—
2AA	GND	GND	Under any condition		Below 1.0	• Related wiring harness
2AB	—	—	—		—	—
2AC	—	—	—		—	—
2AD	GND	Sensor shield	Under any condition		Below 1.0	• Related wiring harness
2AE*10	Fuel pump control	Fuel pump control module	(See Fuel pump control signal.)			• Fuel pump control module • Related wiring harness
2AF*9	A/C cut-off control	A/C relay	A/C relay OFF		B+	• A/C relay • Related wiring harness
			A/C relay ON		Below 1.0	
2AG	HO2S (–)	HO2S	Idle (after warm up)		Approx. 1.63	• HO2S • Related wiring harness
2AH*6	GND	Power brake unit vacuum sensor, clutch stroke sensor*7	Under any condition		Below 1.0	• Related wiring harness
2AI	HO2S (+)	HO2S	Idle (after warm up)		Approx. 2.35	• HO2S • Related wiring harness
2AJ	GND	Refrigerant pressure sensor, ambient temperature sensor	Under any condition		Below 1.0	• Related wiring harness
2AK	HS CAN_H	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness

Terminal	Signal	Connected to	Test condition		Voltage (V)	inspection item
2AL	HS CAN_L	CAN system related modules	Because this terminal is for CAN, integrity determination by terminal voltage is not possible.			• Related wiring harness
2AM*10	Fuel pump control (diagnostic)	Fuel pump control module	(See Fuel pump control module (diagnostic) signal.)			• Fuel pump control module • Related wiring harness
2AN	APP (No.1)	APP sensor No.1	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 0.82	• APP sensor No.1 • Related wiring harness
				Accelerator pedal fully depressed	Approx. 4.58	
2AO	GND	APP sensor No.1	Under any condition		Below 1.0	• Related wiring harness
2AP	—	—	—		—	—
2AQ	Fuel pump control	Fuel pump relay	Ignition switched ON (engine off)		B+	• Fuel pump relay • Related wiring harness
			Idle (after warm up)		Below 1.0	
2AR	Constant voltage (Vref)	APP sensor No.1	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
2AS	APP (No.2)	APP sensor No.2	Ignition switched ON (engine off)	Accelerator pedal released	Approx. 0.42	• APP sensor No.2 • Related wiring harness
				Accelerator pedal fully depressed	Approx. 2.30	
2AT	GND	APP sensor No.2	Under any condition		Below 1.0	• Related wiring harness
2AU	Cooling fan control	Cooling fan relay No.2	Cooling fan operating		Below 1.0	• Cooling fan relay No.2 • Related wiring harness
			Cooling fan not operating		B+	
2AV	Cooling fan control	Cooling fan relay No.1	Cooling fan operating		Below 1.0	• Cooling fan relay No.1 • Related wiring harness
			Cooling fan not operating		B+	
2AW	Constant voltage (Vref)	APP sensor No.2	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness
2AX*9	Refrigerant pressure	Refrigerant pressure sensor	Refrigerant pressure: 1.0 MPa {10 kgf/cm^2, 145 psi}		Approx. 1.58	• Refrigerant pressure sensor • Related wiring harness
			Refrigerant pressure: 1.1 MPa {11 kgf/cm^2, 160 psi}		Approx. 1.72	
			Refrigerant pressure: 1.2 MPa {12 kgf/cm^2, 174 psi}		Approx. 1.88	
2AY	GND	MAF sensor, IAT sensor No.1	Under any condition		Below 1.0	• Related wiring harness
2AZ	Starter cut-off control	Starter relay, start stop unit	Ignition switched ON (engine off)	**MTX** • Clutch pedal released **ATX** • Selector lever position is not P or N position	B+	• Starter relay • Start stop unit • Related wiring harness
				MTX • Clutch pedal fully depressed **ATX** • Selector lever position is P or N position	Below 1.0	
2BA	—	—	—		—	—
2BB	Constant voltage (Vref)	Refrigerant pressure sensor*9, MAF sensor	Ignition switched ON (engine off)		Approx. 5.02	• Related wiring harness

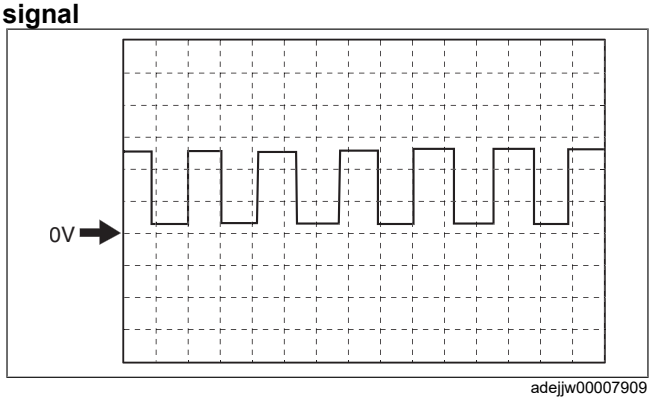
Terminal	Signal	Connected to	Test condition	Voltage (V)	inspection item
2BC	MAF	MAF sensor	Ignition switched ON (engine off)	Approx. 0.76	• MAF sensor • Related wiring harness
			Idle (after warm up)	Approx. 1.12	
			Racing (Engine speed: 2,000 rpm)	Approx. 1.44	
2BD	Selector lever position*1	TCM, start stop unit	Selector lever position is not P or N position	B+	• TCM • Start stop unit • Related wiring harness
			Selector lever position is P or N position	Below 1.0	
	Starter interlock*2	Starter interlock switch, start stop unit	Clutch pedal fully depressed	Below 1.0	• Starter interlock switch • Start stop unit • Related wiring harness
			Clutch pedal released	B+	
2BE	—	—	—	—	—
2BF	—	—	—	—	—
2BG*6	Constant voltage (Vref)	Power brake unit vacuum sensor, clutch stroke sensor*2	Switch ignition ON (engine off)	Approx. 5.02	• Related wiring harness
2BH	—	—	—	—	—

*1 : ATX
*2 : MTX
*3 : With i-ELOOP
*4 : Without i-ELOOP
*5 : SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
*6 : With i-stop system
*7 : With i-stop system (MTX)
*8 : With i-stop system and without i-ELOOP
*9 : With air conditioner
*10 : With fuel pump control module

Inspection Using An Oscilloscope (Reference)
Electric variable valve timing motor (rotation direction) signal
PCM terminals
- 1J(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

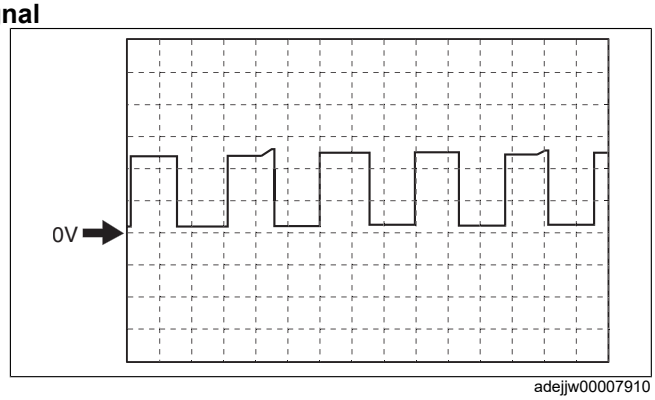


adejjw00007909

Electric variable valve timing motor (rotation pulse) signal
PCM terminals
- 1O(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

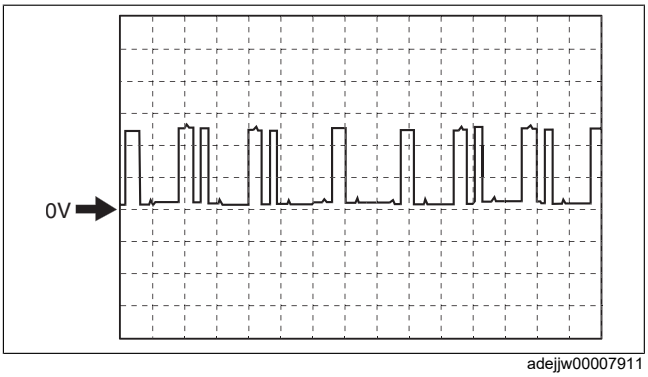


adejjw00007910

Exhaust CMP signal
PCM terminals
- 1T(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

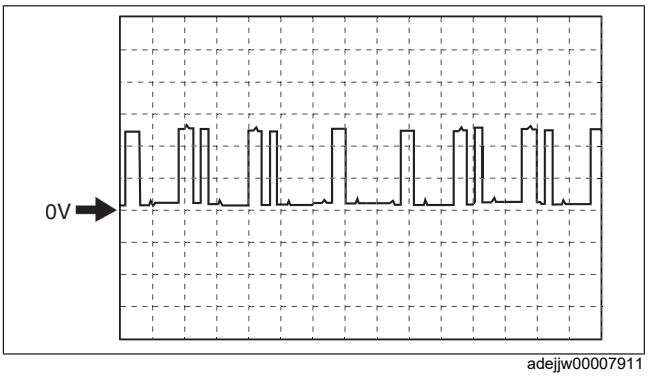


adejjw00007911

Intake CMP signal
PCM terminals
- 1Y(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 20 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

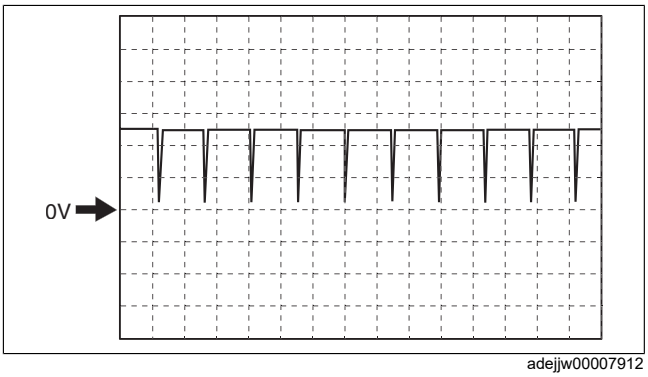


adejjw00007911

CKP signal
PCM terminals
- 1AD(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

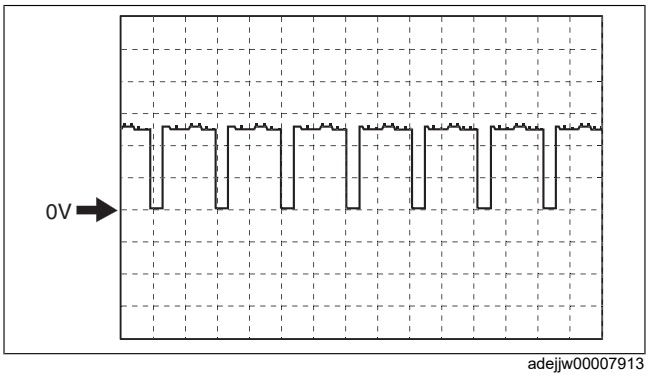


adejjw00007912

Electric variable valve timing driver (diagnostic) signal
PCM terminals
- 1AE(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 100 ms/DIV (X), DC range

Vehicle condition
- Idle (after warm up)

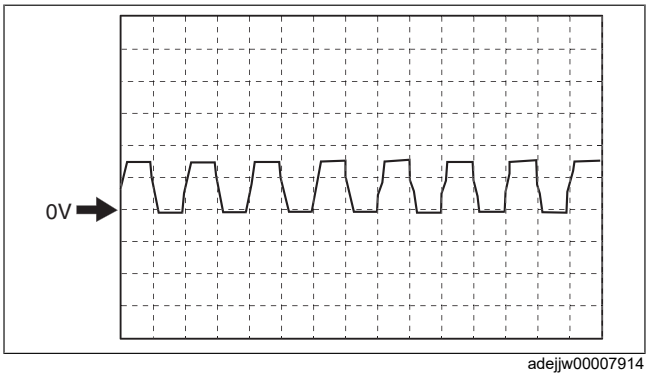


adejjw00007913

Generator output voltage (without i-ELOOP)
PCM terminals
- 1AF(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

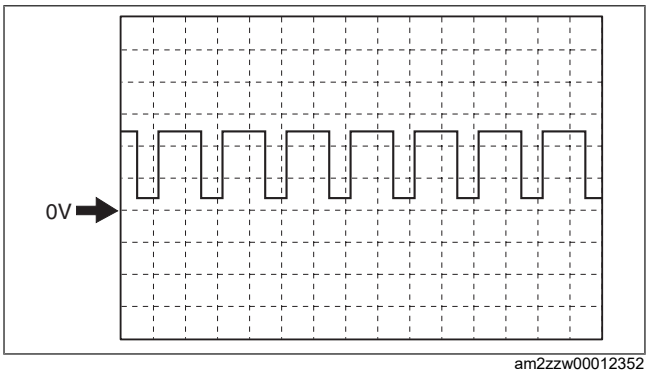


adejjw00007914

Generator output voltage (with i-ELOOP)
PCM terminals
- 1Z(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

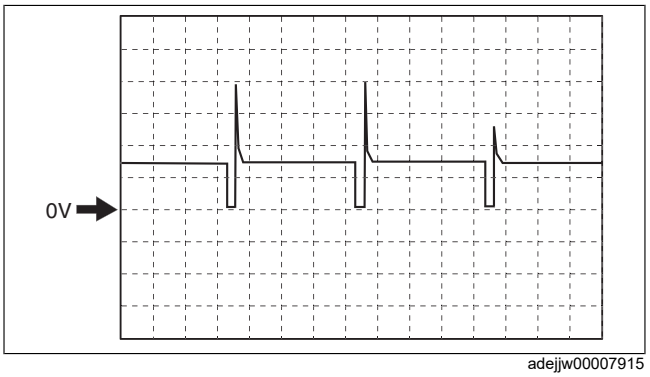


am2zzw00012352

Purge control
PCM terminals
- 1AI(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

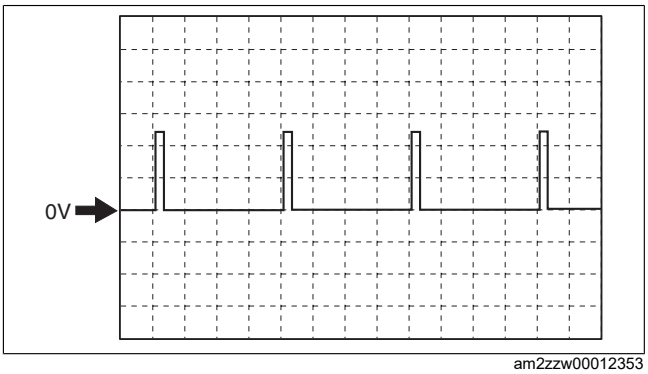


adejjw00007915

IGT1, IGT2, IGT3, IGT4 control
PCM terminals
- IGT1 (ignition coil No.1): 1AY(+)—body ground(–)
- IGT2 (ignition coil No.2): 1AT(+)—body ground(–)
- IGT3 (ignition coil No.3): 1AO(+)—body ground(–)
- IGT4 (ignition coil No.4): 1AJ(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

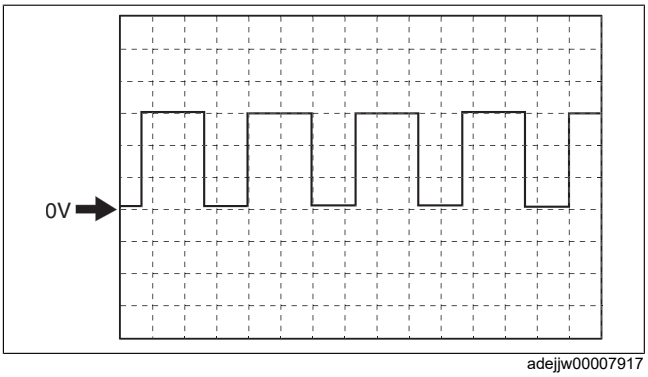


am2zzw00012353

Hydraulic variable valve timing control signal
PCM terminals
- 1AN(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Racing (Engine speed: 2,000 rpm)

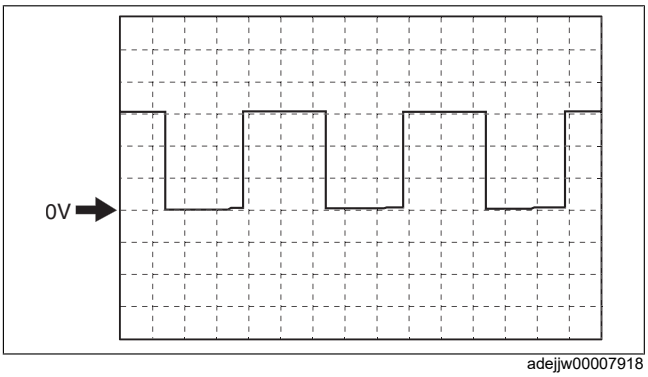


adejjw00007917

Engine oil control signal
PCM terminals
- 1AS(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

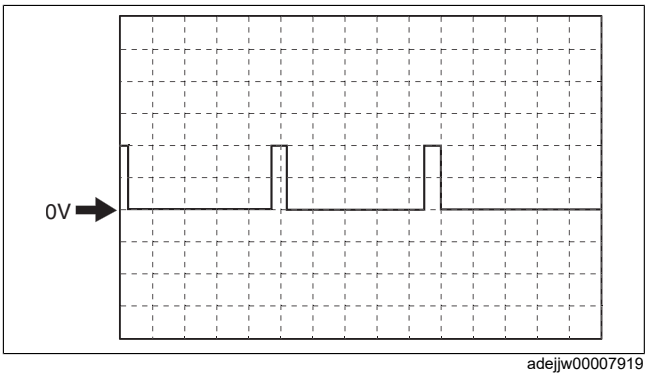


adejjw00007918

Electric variable valve timing control signal
PCM terminals
- 1AZ(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

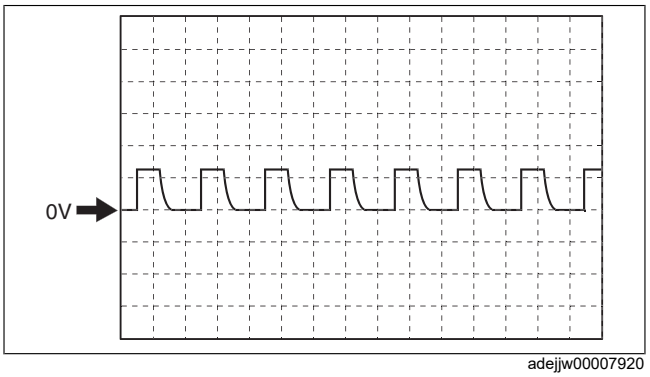


adejjw00007919

Generator field coil control signal (without i-ELOOP)
PCM terminals
- 1BE(+)—body ground(–)

Oscilloscope setting
- 1 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

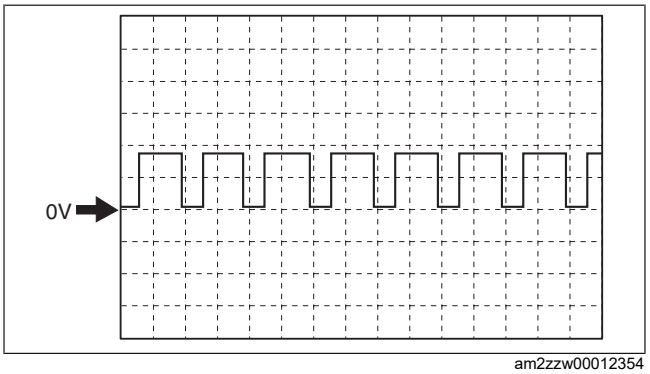


adejjw00007920

Generator field coil control signal (with i-ELOOP)
PCM terminals
- 1BR(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

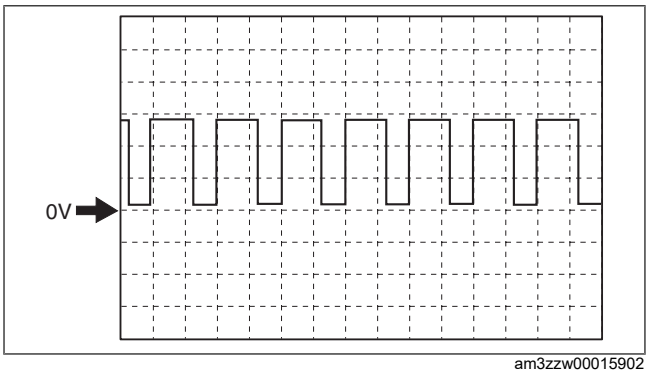


am2zzw00012354

A/F sensor heater control signal
PCM terminals
- 1BY(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



am3zzw00015902

Drive-by-wire control (+) signal

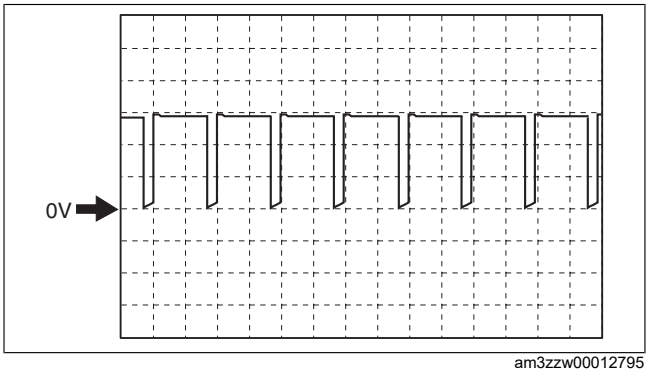
Note
- The wave pattern is either type A or B depending on the PCM specifications.

Type A

PCM terminals
- 1CG(+)—1CC(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



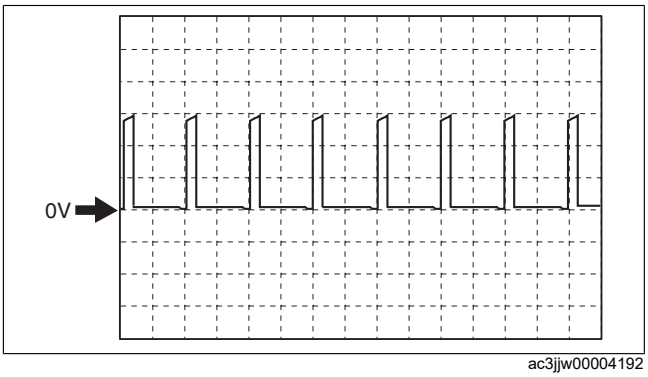
am3zzw00012795

Type B

PCM terminals
- 1CG(+)—1CC(–)

Oscilloscope setting
- 5 V/DIV (Y), 1 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



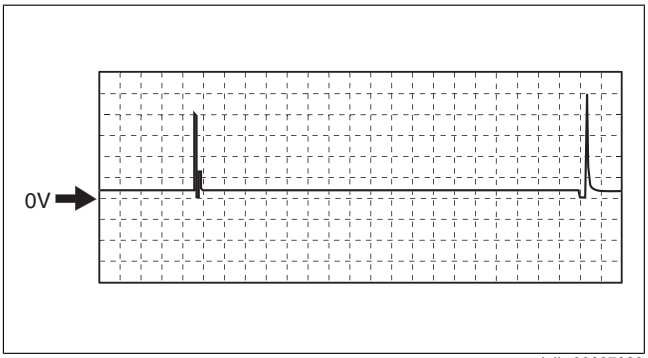
ac3jjw00004192

Fuel injection control (-) signal

PCM terminals
- Fuel Injection No.1: 1DO(+)—body ground(–)
- Fuel Injection No.2: 1DW(+)—body ground(–)
- Fuel Injection No.3: 1EA(+)—body ground(–)
- Fuel Injection No.4: 1DS(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



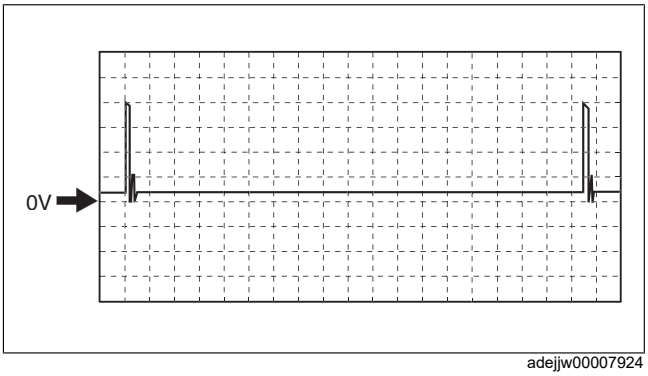
adejjw00007923

Fuel injection control (+) signal

PCM terminals
- Fuel Injection No.1: 1DP(+)—body ground(–)
- Fuel Injection No.2: 1DX(+)—body ground(–)
- Fuel Injection No.3: 1EB(+)—body ground(–)
- Fuel Injection No.4: 1DT(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



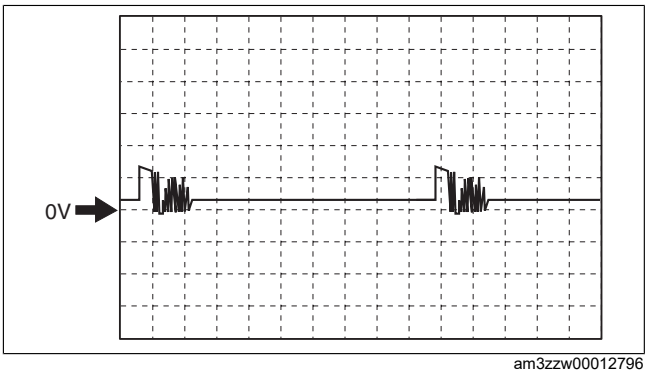
adejjw00007924

High pressure fuel pump control (+) signal

PCM terminals
- 1EE(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

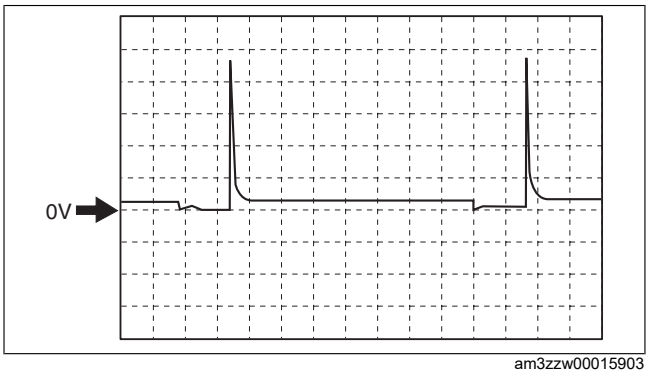


am3zzw00012796

High pressure fuel pump control (-) signal
PCM terminals
- 1EF(+)—body ground(–)

Oscilloscope setting
- 10 V/DIV (Y), 5 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

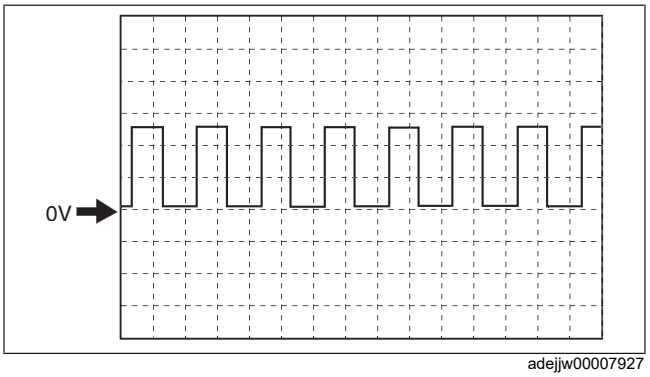


am3zzw00015903

HO2S heater control signal
PCM terminals
- 2C(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition
- Idle (immediately after starting engine)

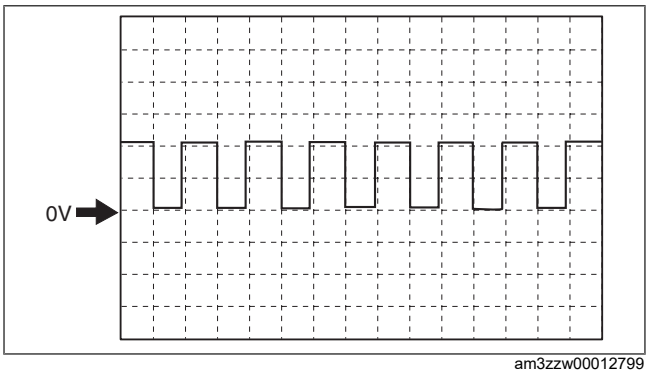


adejjw00007927

Fuel pump control signal
PCM terminals
- 2AE(+)—body ground(–)

Oscilloscope setting
- 5 V/DIV (Y), 2 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up

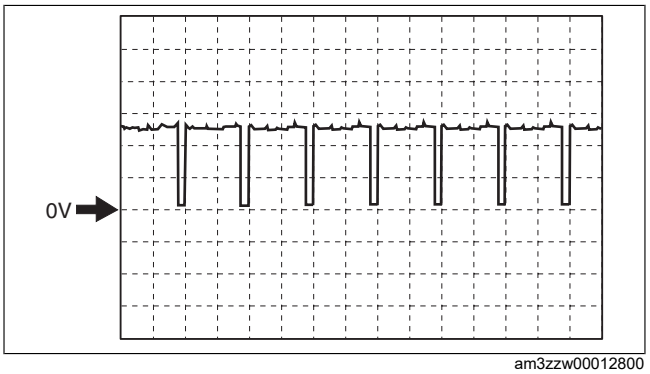


am3zzw00012799

Fuel pump control module (diagnostic) signal
PCM terminals
- 2AM(+)—body ground(–)

Oscilloscope setting
- 2 V/DIV (Y), 50 ms/DIV (X), DC range

Vehicle condition
- Idle after warm up



am3zzw00012800

Using The M-MDS

Note
- PIDs for the following parts are not available on this model. Go to the appropriate part inspection page.
  - — Intake CMP sensor and exhaust CMP sensor (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

— Main relay (See RELAY INSPECTION.)

1. Connect the **M-MDS** to the DLC-2.
2. Switch the ignition ON (engine off).
3. Measure the PID value.
   - If PID value is not within the specification, follow the instructions in action column.

Note
- The PID/DATA MONITOR function monitors the calculated value of the input/output signals in the PCM. Therefore, an output device malfunction is not directly indicated as a malfunction of the monitored value for the output device. If a monitored value of an output device is out of specification, inspect the monitored value of the input device related to the output control.
- The simulation items that are used in the engine control system operation inspection are as follows.
  — ACCS, ARPMDES, EVAPCP, FAN1, FAN3, FP, INJ_1, INJ_2, INJ_3, INJ_4, OIL_P_SOL

—: Not applicable

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
AC_PRES	KPa {MPa}, mBar {Bar}, psi, in H20	Refrigerant pressure input from refrigerant pressure sensor	Calculation	• Displays refrigerant pressure	2AX
	V	Refrigerant pressure sensor voltage	Input	• Refrigerant pressure is 449 kPa {4.58 kgf/cm^2, 65.1 psi}: Approx. 0.76 V • Refrigerant pressure is 729 kPa {7.43 kgf/cm^2, 106 psi}: Approx. 1.18 V	
AC_REQ	Off/On	A/C switch status received by PCM via CAN	Input	• A/C switch off: Off • A/C switch on: On	CAN (2AK, 2AL)
ACCS	Off/On	A/C relay status input from A/C relay	Input	• A/C relay off: Off • A/C relay on: On	2AF
ALTF	%	Field coil current control signal output to generator	Calculation	• Ignition switched ON (engine off): 0% • Idle: Approx. 41—54% • Racing (Engine speed is 2,000 rpm): Approx. 24—34%	1BR*5, 1BE*6
ALTF_ACT*5	%	Actually measured value of field coil current signal input from generator	Calculation	• Displays actual generator field current control duty value	—
ALTT V*6	V	Generator output voltage	Input	• Idle (no E/L): Approx. 14 V (This is internal calculation value and differs from terminal voltage)	1AF
AMB_TEMP	°C, °F	Actually measured ambient temperature input from ambient temperature sensor	Calculation	• Displays ambient air temperature	2I
APP	%	Accelerator pedal opening angle (relative value) with the fully released status as 0% and fully depressed status as 100%	Calculation	• Accelerator pedal released: Approx. 0% • Accelerator pedal fully depressed: Approx. 100%	—
APP1	%	Accelerator pedal opening angle (absolute value) input from APP sensor No. 1	Calculation	• Accelerator pedal released: Approx. 16% • Accelerator pedal fully depressed: Approx. 91%	2AN
	V	APP sensor No.1 voltage	Input	• Accelerator pedal released: Approx. 0.78 V • Accelerator pedal fully depressed: Approx. 4.54 V	

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
APP2	%	Accelerator pedal opening angle (absolute value) input from APP sensor No. 2	Calculation	• Accelerator pedal released: Approx. 7.84% • Accelerator pedal fully depressed: Approx. 45.49%	2AS
	V	APP sensor No.2 voltage	Input	• Accelerator pedal released: Approx. 0.39 V • Accelerator pedal fully depressed: Approx. 2.27 V	
ARPMDES	RPM	Target engine speed	Calculation	• Displays target engine speed	—
BARO	KPa {MPa}, mBar {Bar}, psi, in H20	Actually measured barometric pressure input from barometric pressure sensor built into PCM	Calculation	• Displays BARO	—
BATT_CUR*7	A	Battery charge/discharge current	Calculation	• Displays battery charge/discharge current value	CAN (2AK, 2AL)
BATT_DAY*7	—	Number of days elapsed since current sensor initialization	Calculation	• Displays vehicle battery days in service	CAN (2AK, 2AL)
BATT_RES*7	—	Battery internal resistance (estimated)	Calculation	• Displays battery inferred internal resistance	CAN (2AK, 2AL)
BATT_SOC*7	%	Battery charge condition (estimated)	Calculation	• Displays battery estimated state of charge	CAN (2AK, 2AL)
BATT_TEMP*7	°C, °F	Battery temperature	Calculation	• Displays battery fluid temperature	CAN (2AK, 2AL)
BATT_V*7	V	Battery voltage	Calculation	• Displays battery voltage	CAN (2AK, 2AL)
BBP*3	KPa {MPa}, mBar {Bar}, psi, in H20	Power brake unit vacuum input from power brake unit vacuum sensor	Calculation	• Displays power brake unit vacuum	2Q
	V	Power brake unit vacuum sensor voltage	Input	**Ignition switched ON** • Power brake unit vacuum is 7.96 kPa {0.0812 kgf/cm^2, 1.15 psi}: approx. 0.31 V • Power brake unit vacuum is 97.23 kPa {0.9915 kgf/cm^2, 14.10 psi}: approx. 3.93 V	
BFP*3	KPa {MPa}, mBar {Bar}, psi, in H20	Actually measured brake fluid pressure input from brake fluid pressure sensor built into DSC HU/ CM via CAN	Calculation	• Displays brake fluid pressure	CAN (2AK, 2AL)
BOO	High/Low	Brake switch (No.1 signal) input status	Calculation	• Brake pedal released: Low • Brake pedal fully depressed: High	2G
BPA	High/Low	Brake switch (No.2 signal) input status	Calculation	• Brake pedal released: Low • Brake pedal fully depressed: High	2R
CATT11_DSD	°C, °F	Estimated catalytic converter temperature	Calculation	• Displays estimated catalytic converter temperature **Idle (after warm up)** — ECT is 88 °C {190 °F}: Approx. 304 °C {579 °F}	—
CHRGLP	Off/On	Charging system warning light illumination status	Calculation	• Charging system warning light not illuminated: Off • Charging system warning light illuminated: On	—
CLU_CUT_SW *4	Off/On	Starter interlock switch status	Calculation	• Clutch pedal released: Off (starter interlock switch OFF) • Clutch pedal fully depressed: On (starter interlock switch ON)	2BD
CPP*1	Off/On	Clutch pedal position	Calculation	• Clutch pedal released: Off • Clutch pedal fully depressed: On	2J

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
CPP*4	%	Clutch pedal position input from clutch stroke sensor	Calculation	• Clutch pedal released: approx. 0.2% • Clutch pedal fully depressed: approx. 93.5%	2M
CPP/PNP*1	Off/On	Shift lever position	Calculation	• Other than neutral: Off • Neutral: On	1K
ECT	°C, °F	Engine coolant temperature input from ECT sensor	Calculation	• Displays ECT	1AK
	V	ECT sensor voltage	Input	**Ignition switched ON (engine off)** • ECT is 29 °C {84 °F}: Approx. 2.65 V **Idle (after warm up)** • ECT is 88 °C {190 °F}: Approx. 0.71 V	
EQ_RAT11	—	Excess air factor (estimated value) to theoretical air/fuel ratio (14.7) by fuel feedback control	Calculation	**Idle (after warm up)** • ECT is 89 °C {192 °F}: Approx. 1	—
EQ_RAT11_DSD	—	Target excess air factor (estimated value) to theoretical air/fuel ratio (14.7) by fuel feedback control	Calculation	• Indicate target lambda (Excess air factor = supplied air amount / theoretical air/fuel ratio)	—
ETC_ACT	°	Actual throttle valve opening angle	Calculation	**Ignition switched ON (engine off)** • Accelerator pedal released: Approx. 12.89 ° • Accelerator pedal fully depressed: Approx. 86.03 ° **Idle (after warm up)** • Accelerator pedal released: Approx. 2.77—2.84 ° (ECT is 89 °C {192 °F})	—
ETC_DSD	°	Target throttle valve opening angle	Calculation	• Displays target TP opening angle **Ignition switched ON (engine off)** — Accelerator pedal released: Approx. 12.9 ° — Accelerator pedal fully depressed: Approx. 86 °	—
	%	Target throttle valve opening angle (percent)	Calculation	• Displays target TP opening angle (percent) **Ignition switched ON (engine off)** — Accelerator pedal released: Approx. 14.9% — Accelerator pedal fully depressed: Approx. 100 %	
EVAPCP	%	Purge solenoid valve control duty value	Calculation	• Idle (after warm up): 0% (Engine coolant temperature 59 °C {140 °F} or less) • Racing (Engine speed 2,000 rpm): 33—35% (ECT is 94 °C {201 °F}) • Racing (Engine speed 4,000 rpm): 53—55% (ECT is 94 °C {201 °F})	1AI
FAN1	Off/On	Cooling fan relay No.2 operation status	Calculation	• Cooling fan relay No.2 off: Off • Cooling fan relay No.2 on: On	2AU
FAN3	Off/On	Cooling fan relay No.1 operation status	Calculation	• Cooling fan relay No.1 off: Off • Cooling fan relay No.1 on: On	2AV
FIA	—	Fuel injection amount	Calculation	• Displays fuel injection amount **Idle (after warm up)** — ECT is 90 °C {194 °F}: Approx. 6	—
FLI	%	Fuel level	Calculation	• Fuel gauge level F: Approx. 100% • Fuel gauge level E: Approx. 0%	CAN (2AK, 2AL)
FP	Off/On	Fuel pump relay operation status	Calculation	• Ignition switched ON (engine off): Off • Cranking: On • Idle (after warm up): On	2AQ

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
FP_DUTY*8	%	Fuel pump control module control duty value	Calculation	• Ignition switched ON (engine off): Approx. 55.74% • Cranking: Approx. 95% • Idle (after warm up): Approx. 55.74%	2AE
FUEL_P_DSD	KPa {MPa}, mBar {Bar}, psi, in H20	Target fuel pressure (high pressure fuel pump)	Calculation	• Displays target fuel pressure (high pressure fuel pump) **Ignition switched ON (engine off)** — Fuel pressure: Approx. 12 MPa {122 kgf/cm^2, 1740 psi} **Idle (after warm up)** — ECT is 91 °C {196 °F}: Fuel pressure is 3.3—3.8 MPa {34—38 kgf/cm^2, 479—551 psi}	—
FUEL_PRES	KPa {MPa}, mBar {Bar}, psi, in H20	Fuel pressure input from fuel pressure sensor	Calculation	• Displays fuel pressure	1CA
	V	Fuel pressure sensor voltage	Input	**Ignition switched ON (engine off)** • Fuel pressure is approx. 16.91 MPa {172.4 kgf/cm^2, 2453 psi}: Approx. 2.91 V **Idle (after warm up)** • Fuel pressure is 3.44—3.95 MPa {35.1—40.2 kgf/cm^2, 499—572 psi}: 0.98—1.05 V (ECT is 90 °C {194 °F})	
FUELPW	Sec	Fuel injection pulse width (fuel injector energization time) output to fuel injector	Calculation	• Idle (after warm up): Approx. 1.71 ms • Racing (engine speed is 2,000 rpm): Approx. 1.35 ms • Racing (engine speed is 4,000 rpm): Approx. 1.21 ms	No.1:1DO/ 1DP No.2:1DW/ 1DX No.3:1EA/ 1EB No.4:1DS/ 1DT
FUELSYS	OL/CL/ OL_Drive/ OL_Fault/ CL_Fault	Feedback status of fuel injection control is displayed OL: Feedback control is disabled at cold engine start CL: During feedback control by A/F sensor and HO2S OL_Drive: While feedback control is stopped OL_Fault: Feedback control is disable due to system malfunction CL_Fault: During feedback control with either A/F sensor or HO2S having a malfunction	Calculation	• Idle (after warm up): OL or CL • Racing (engine speed is 2,000 rpm): CL • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): OL-Drive	—
GEAR*2	Unknown/ 1st/2nd/3rd/ 4th/5th/6th/ Not in P/ Park/ Neutral/ Drive/ Reverse	Gear commanded	Calculation	• Selector lever at P position: Park • Selector lever at R position: Reverse • Selector lever at N position: Neutral • Selector lever is in D or M position while vehicle is stopped: 1st	CAN (2AK, 2AL)
HTR11	Off/On	A/F sensor heater operation status	Calculation	• Ignition switched ON (engine off): Off • Idle (after warm up): On	1BY
	%	A/F sensor heater control duty value	Calculation	• Ignition switched ON (engine off): 0% • Idle (after warm up): 32—34%	

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
HTR12	Off/On	HO2S heater operation status	Calculation	• Ignition switched ON (engine off): Off • Idle (after warm up): On	2C
	%	HO2S heater control duty value	Calculation	• Ignition switched ON (engine off): 0% • Idle (after warm up): 41—48%	
IAT	°C, °F	Intake air temperature (No.1) input from IAT sensor No.1	Calculation	• Displays IAT (No.1)	2U
	V	IAT sensor No.1 voltage	Input	• IAT is 20 °C {68 °F}: Approx. 2.70 V • IAT is 40 °C {104 °F}: Approx. 1.80 V • IAT is 60 °C {140 °F}: Approx. 1.20 V	
IAT2	V	IAT sensor No.2 voltage	Input	• IAT2 is 20 °C {68 °F}: Approx. 3.57 V • IAT2 is 40 °C {104 °F}: Approx. 2.70 V • IAT2 is 60 °C {140 °F}: Approx. 1.87 V	1CE
INGEAR	Off/On	Gears are engaged	Calculation	**MTX** • When the following conditions are satisfied: On — Other than neutral — Clutch pedal released • Except above: Off **ATX** • Selector lever is in R, D, or M position: On • Except above: Off	CAN (2AK, 2AL)
ISC_FBK	%	ISC feedback value	Calculation	• Displays ISC feedback value	—
I-Stop_OFF*3	Off/On	i-stop OFF mode	Calculation	• i-stop OFF switch OFF: Off • i-stop OFF switch ON: On	CAN (2AK, 2AL)
I-Stop_TRD*3	Off/On	i-stop transmission D position selected status	Calculation	• D position:On • Except above: Off	CAN (2AK, 2AL)
I-Stop_VSP*3	Off/On	i-stop vehicle speed history flag	Calculation	• Vehicle speed in which engine stop condition is met via i-stop control is detected: On • Except above: Off	CAN (2AK, 2AL)
I-Stop_VST*3	Off/On	i-stop vehicle stop flag	Calculation	• Vehicle stop predicted: On • Except above: Off	CAN (2AK, 2AL)
IVS	Off Idle/Idle	Idle flag	Calculation	• Racing: Off Idle • Idle: Idle	—
KNOCKR	°	Ignition timing correction for suppressing engine knock (Performs retard correction (negative indication) according to the occurrence of engine knock, and it approaches approx. 0° by the advance correction due to engine knock suppression.)	Calculation	• Ignition switched ON (engine off): 0 ° • Idle (after warm up): 0 °	—
LOAD	%	Ratio of actual amount of intake air to the maximum air charging amount (mass volume) of cylinder	Calculation	• Idle (after warm up): Approx. 22.35% • Racing (engine speed is 2,000 rpm): Approx. 17.25% • Racing (engine speed is 4,000 rpm): Approx. 19.21%	—
LONGFT1	%	Fuel learning correction amount estimated based on A/F sensor signal	Calculation	• Idle (after warm up): Approx. −6.25% • Racing (engine speed is 2,000 rpm): Approx. −3.9% • Racing (engine speed is 4,000 rpm): Approx. −3.9%	—
LONGFT12	%	Fuel learning correction amount estimated based on HO2S signal	Calculation	• Idle (after warm up): 0%	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
LOW_OIL	Never Detected/ Detected	Engine oil pressure condition • Never Detected: Engine oil pressure is high • Detected: Engine oil pressure is low	Calculation	• Ignition switched ON (engine off): Detected • Idle (after warm up): Detected • Racing (engine speed is 2,000—4,000 rpm): Never Detected	1P
LRN_KCS	—	Knock control system learning value	Calculation	• Displays knock control system learning value	—
M_GEAR*1	Neutral/1st gear/2nd gear/3rd gear/4th gear/5th gear/6th gear/ Reverse/ Undefined/ Auto/ In_Progress /YSF/Error	Manual gear position	Calculation	• Displays manual gear position	CAN (2AK, 2AL)
MAF	g/Sec	Mass air flow input from MAF sensor	Calculation	• Displays MAF	2BC
	V	MAF sensor voltage	Input	• Ignition switched ON (engine off) (MAF: 0.59 g/s {0.078 lb/min}): Approx. 0.72 V • Idle (after warm up) (MAF: 2.17 g/s {0.287 lb/min}): Approx. 0.97 V • Racing (engine speed is 2,000 rpm) (MAF: 4.73 g/s {0.626 lb/min}): Approx. 1.26 V	
MAP	KPa {MPa}, mBar {Bar}, psi, in H20	Manifold absolute pressure input from MAP sensor	Calculation	• Displays MAP	1BW
MAP_V	V	MAP sensor voltage	Input	• Ignition switched ON (engine off) (MAP:101 kPa {1.03 kgf/cm^2, 14.6 psi}): Approx. 4.12 V • Idle (after warm up) (MAP: 33 kPa {0.34 kgf/cm^2, 4.8 psi}): Approx. 1.9 V • Racing (engine speed is 2,000 rpm) (MAP: 26 kPa {0.27 kgf/cm^2, 3.8 psi}): Approx. 1.23 V	
MF_CAT1	—	Number of misfires in No.1 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.1 cylinder)	—
MF_CAT_2	—	Number of misfires in No.2 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.2 cylinder)	—

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
MF_CAT_3	—	Number of misfires in No.3 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.3 cylinder)	—
MF_CAT_4	—	Number of misfires in No.4 cylinder leading to catalytic converter temperature increase (catalytic converter temperature increases due to fuel combustion around catalytic converter after misfire)	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (No.4 cylinder)	—
MF_CAT_FCC	—	Threshold of malfunction determination for number of misfires (total number in all cylinders) leading to catalytic converter temperature increase	Calculation	• Displays number of misfire determinations (for catalytic converter)	—
MF_CAT_TTL	—	Number of misfires (total number in all cylinders) leading to catalytic converter temperature increase	Calculation	• Displays number of misfires corresponding to possible catalytic converter damage (total)	—
MF_EMI1	—	Number of misfires in No.1 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.1 cylinder)	—
MF_EMI_2	—	Number of misfires in No.2 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.2 cylinder)	—
MF_EMI_3	—	Number of misfires in No.3 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.3 cylinder)	—
MF_EMI_4	—	Number of misfires in No.4 cylinder under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (No.4 cylinder)	—
MF_EMI_FCC	—	Threshold of malfunction determination for number of misfires (total number in all cylinders) under conditions required by emission regulations	Calculation	• Displays number of misfire determinations (for emission)	—
MF_EMI_TTL	—	Number of misfires (total number in all cylinders) under conditions required by emission regulations	Calculation	• Displays number of misfires possibly affecting emission (total)	—
MIL	Off/On	Check engine light illumination status	Calculation	• Check engine light not illuminated: Off • Check engine light illuminated: On	—
MIL_DIS	km, ft, mi	Travelled distance since check engine light illuminated	Calculation	• Displays travelled distance since check engine light illuminated	—
NEUTRAL_SW 1*4	Off/On	Neutral switch No.1 status	Calculation	• Neutral: On • Other than neutral: Off	1K

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
NEUTRAL_SW2*4	Off/On	Neutral switch No.2 status	Calculation	• Neutral: On • Other than neutral: Off	1G
O2S11	μA	A/F sensor current	Input	• Idle (after warm up): 0—50 μA • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 3.84 mA	1AB
O2S12	V	HO2S voltage	Input	• Idle (after warm up): 0—1.0 V • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 0 V	2AI/2AG
OIL_P_SOL	Off/On	Engine oil solenoid valve operation status	Calculation	• ECT below 98 °C {208 °F} and engine speed below 4,000 rpm: On • ECT above 98 °C {208 °F} or engine speed above 4,000 rpm: Off	1AS
OIL_TEMP	°C, °F	Estimated engine oil temperature	Calculation	• Displays estimated engine oil temperature	—
PN_SW*2	Open/ Closed	Parking/neutral	Calculation	• Selector lever at P position or N position: Closed • Except above: Open	CAN (2AK, 2AL)
RO2FT1	%	Fuel correction amount based on HO2S signal including SHRTFT12 and LONGFT12	Calculation	• Idle (after warm up): Approx. 0% • Deceleration fuel cut (accelerator pedal released from engine speed of 4,000 rpm or more): Approx. 3.99%	—
RPM	RPM	Engine speed	Calculation	• Displays engine speed	1AD
SHRTFT1	%	Fuel feedback correction amount estimated based on A/F sensor signal	Calculation	• Idle (after warm up): Approx. 0% • Racing (engine speed is 2,000 rpm): Approx. 1.56% • Racing (engine speed is 4,000 rpm): Approx. −4.68%	—
SHRTFT12	%	Fuel feedback correction amount estimated based on HO2S signal	Calculation	• Idle (after warm up): 0%	—
SPARKADV	°	Ignition timing	Calculation	• Displays ignition timing	—
TP_REL	%	Throttle valve opening angle (relative value) with value at throttle valve fully close timing as the start point	Calculation	• Accelerator pedal released: Approx. 12% • Accelerator pedal fully depressed: Approx. 82%	—
TP1	%	Throttle valve position No. 1	Calculation	• Accelerator pedal released: Approx. 22% • Accelerator pedal fully depressed: Approx. 92%	1BP
	V	TP sensor No.1 voltage	Input	• Accelerator pedal released: Approx. 1.11 V • Accelerator pedal fully depressed: Approx. 4.59 V	
TP2	%	Throttle valve position No. 2	Calculation	• Accelerator pedal released: Approx. 22% • Accelerator pedal fully depressed: Approx. 92%	1BU
	V	TP sensor No.2 voltage	Input	• Accelerator pedal released: Approx. 3.92 V • Accelerator pedal fully depressed: Approx. 0.41 V	
TPCT	V	TP sensor No.1 minimum voltage at CTP	Calculation	• Ignition switched ON (engine off): Approx. 0.5 V	1BP
TPCT2	V	TP sensor No.2 maximum voltage at CTP	Calculation	• Ignition switched ON (engine off): Approx. 4.5 V	1BU
VPWR	V	Battery positive voltage	Input	• Displays battery voltage	2O
VSS	KPH, MPH	Vehicle speed	Calculation	• Displays vehicle speed	CAN (2AK, 2AL)

Item (definition)	Unit/ Condition	Definition	Value type	Condition/Specification (Reference)	PCM terminal
VT_EX_DES	°	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Desired intake valve timing **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Desired exhaust valve timing	Calculation	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Displays desired intake valve timing **Idle (after warm up)** — Approx. 0 ° **Racing (engine speed is 2,000 rpm)** — Approx. 26 ° **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Displays desired exhaust valve timing	—
VT_IN_ACT	°	Actual intake valve timing	Calculation	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Displays in the M-MDS but it does not operate. **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Displays actual intake valve timing **Idle (after warm up)** — Approx. 57 ° **Racing (engine speed is 2,000 rpm)** — Approx. 84 °	1Y*9
VT_IN_DES	°	Desired intake valve timing	Calculation	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Displays in the M-MDS but it does not operate. **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Displays desired intake valve timing **Idle (after warm up)** — Approx. 0 ° **Racing (engine speed is 2,000 rpm)** — Approx. 29 °	—
VT_EX_ACT	°	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Actual intake valve timing **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Actual exhaust valve timing	Calculation	**SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)** • Displays actual intake valve timing **Idle (after warm up)** — Below 1 ° **Racing (engine speed is 2,000 rpm)** — Approx. 26 ° **SKYACTIV-G 1.5 (with 4-2-1 exhaust system)** • Displays actual exhaust valve timing	1Y*8, 1T*9
VT_EX_DUTY	%	OCV control duty value	Calculation	• Idle (after warm up): 0% • Racing (engine speed is 2,000 rpm): Approx. 40%	1AN

*1 : MTX
*2 : ATX
*3 : With i-stop system
*4 : With i-stop system (MTX)
*5 : With i-ELOOP
*6 : Without i-ELOOP
*7 : With i-stop or with i-ELOOP
*8 : With fuel pump control module